

Poincaré Recurrence Theorem

Special Mathematics Lectures - Differential equations and dynamical systems

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1 Volume-preserving maps

For a measurable set $\Lambda \subset \mathbb{R}^n$, the volume $\text{vol}(\Lambda)$ of Λ is defined by

$$\text{vol}(\Lambda) = \int_{\Lambda} 1 \, dx = \int \chi_{\Lambda}(x) \, dx,$$

where the integral is defined in term of the Lebesgue measure.

Definition 1. A *volume-preserving map* $\phi : \Omega \rightarrow \mathbb{R}^n$ with $\Omega \subset \mathbb{R}^n$ is a measurable map such that for any measurable set $\Lambda \subset \phi(\Omega)$ with a finite volume $\text{vol}(\Lambda)$,

$$\text{vol}(\phi^{-1}(\Lambda)) = \text{vol}(\Lambda),$$

where $\phi^{-1}(\Lambda)$ is the preimage of Λ under the measurable map ϕ and therefore, it is measurable.

Remark. (i) If a volume-preserving map ϕ satisfies $\phi(\Omega) = \Omega$, then for any $m \geq 2$ and any measurable set $\Lambda \subset \Omega$, we define inductively $\phi^{-m}(\Lambda) := \phi^{-1}(\phi^{-(m-1)}(\Lambda))$ (i.e. we repeatedly take preimage). Then, we have for $m \geq 2$

$$\text{vol}(\phi^{-m}(\Lambda)) = \text{vol}(\phi^{-1}(\phi^{-(m-1)}(\Lambda))) = \text{vol}(\phi^{-(m-1)}(\Lambda)) = \dots = \text{vol}(\phi^{-1}(\Lambda)) = \text{vol}(\Lambda).$$

Therefore, we have $\text{vol}(\phi^{-m}(\Lambda)) = \text{vol}(\Lambda)$ for any $m \in \mathbb{N} = \{1, 2, 3, \dots\}$. This shows that the iterated map ϕ^m for any $m \in \mathbb{N}$ is also volume-preserving in this case.

(ii) If a volume-preserving map ϕ is invertible with a measurable inverse ϕ^{-1} , then ϕ^{-1} is also volume-preserving. Indeed, for any measurable set $\Lambda \subset \Omega$ with a finite volume, we have

$$\text{vol}(\Lambda) = \text{vol}(\phi^{-1}(\phi(\Lambda))) = \text{vol}(\phi(\Lambda)).$$

(iii) If a volume-preserving map ϕ is invertible with a measurable inverse ϕ^{-1} and $\phi(\Omega) = \Omega$, then ϕ^{-1} is volume-preserving by (ii) and we can use the same argument as (i) for ϕ^{-1} to show that $\text{vol}(\phi^m(\Lambda)) = \text{vol}(\Lambda)$ for any $m \in \mathbb{N}$ and measurable set $\Lambda \subset \Omega$. In fact, the map ϕ^m for any $m \in \mathbb{Z}$ is volume-preserving in this case ($\phi^{-m} = (\phi^{-1})^m$ for $m \in \mathbb{N}$ is an iterated map of ϕ^{-1} and $\phi^0 := \text{id}_{\Omega}$).

One particular type of maps that we want to investigate in this report comes from autonomous first-order differential systems.

Definition 2. Consider an autonomous first-order differential system $\dot{x}(t) = f(x(t))$ with a vector field $f \in C^1(\Omega, \mathbb{R}^n)$ for a compact subset $\Omega \subset \mathbb{R}^n$. The *flow* of the system is a family of maps $\{\phi_t : \Omega \rightarrow \mathbb{R}^n\}_{t \in (-\epsilon, \epsilon)}$ for some $\epsilon > 0$ such that $x(t) = \phi_t(x_0)$ for $x_0 \in \Omega$ is a unique solution of the initial value problem

$$\begin{cases} \dot{x}(t) = f(x(t)), \\ x(0) = x_0. \end{cases}$$

Remark. The requirement that f is of class C^1 ensures the existence and uniqueness of solutions $x \in C^2((-\epsilon, \epsilon), \mathbb{R}^n)$ by the Picard-Lindelöf theorem. The flow maps ϕ_t have the following properties:

- (i) $\phi_0 = \text{id}_\Omega$,
- (ii) $\phi_{t+s} = \phi_t \circ \phi_s = \phi_s \circ \phi_t$ for all $t, s \in (-\epsilon, \epsilon)$ such that $t + s \in (-\epsilon, \epsilon)$.
- (iii) Each map ϕ_t for $t \in (-\epsilon, \epsilon)$ is invertible with the inverse ϕ_{-t} since $\phi_t \circ \phi_{-t} = \phi_{-t} \circ \phi_t = \text{id}_\Omega$.

For property (ii) to be valid, we assume that ϵ is sufficiently small and Ω is chosen so that $\phi_t(\Omega) \subseteq \Omega$ for all $t \in (-\epsilon, \epsilon)$, i.e. the orbits do not leave Ω right away. Property (iii) shows that each map ϕ_t is a diffeomorphism from Ω to $\phi_t(\Omega)$ since both ϕ_t and ϕ_{-t} are continuously differentiable. Therefore, ϕ_t is a measurable map with a measurable inverse ϕ_{-t} .

The following proposition provides a sufficient condition for the flow of a differential system to be volume-preserving.

Proposition 3. *Consider an autonomous first-order differential system $\dot{x}(t) = f(x(t))$ with $f \in C^1(\Omega, \mathbb{R}^n)$ for a compact subset $\Omega \subset \mathbb{R}^n$ such that $\text{div}(f) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} = 0$. Then, each map $\phi_t : \Omega \rightarrow \mathbb{R}^n$ of the flow is volume-preserving for all $t \in (-\epsilon, \epsilon)$.*

Proof. For any measurable set $\Lambda \subset \Omega$, the volume of its image under the map ϕ_t is given by

$$\text{vol}(\phi_t(\Lambda)) = \int_{\phi_t(\Lambda)} 1 \, dx = \int_{\Lambda} \det(\mathcal{J}_{\phi_t}(x)) \, dx$$

The [Jacobi's formula](#) for a matrix function $A : \mathbb{R} \rightarrow M_n(\mathbb{R})$ is given by

$$\frac{d}{dt} \det(A) = \det(A) \, \text{tr} \left(A^{-1} \frac{dA}{dt} \right).$$

Using this formula, we have

$$\begin{aligned} \left. \frac{d}{dt} \text{vol}(\phi_t(\Lambda)) \right|_{t=0} &= \left. \frac{d}{dt} \int_{\Lambda} \det(\mathcal{J}_{\phi_t}(x)) \, dx \right|_{t=0} \\ &= \int_{\Lambda} \left. \frac{\partial}{\partial t} \det(\mathcal{J}_{\phi_t}(x)) \right|_{t=0} \, dx \\ &= \int_{\Lambda} \det(\mathcal{J}_{\phi_0}(x)) \, \text{tr} \left(\mathcal{J}_{\phi_0}^{-1}(x) \left. \frac{\partial \mathcal{J}_{\phi_t}(x)}{\partial t} \right|_{t=0} \right) \, dx \\ &= \int_{\Lambda} \text{tr} \left(\left. \frac{\partial \mathcal{J}_{\phi_t}(x)}{\partial t} \right|_{t=0} \right) \, dx \end{aligned}$$

since $\phi_0 = \text{id}_\Omega$ and thus $\mathcal{J}_{\phi_0} = \mathbf{1}$. Interchanging the order of differentiation with respect to t and x_i , we have

$$\left. \frac{\partial (\mathcal{J}_{\phi_t}(x))_{jk}}{\partial t} \right|_{t=0} = \frac{\partial}{\partial t} \left(\left. \frac{\partial (\phi_t(x))_j}{\partial x_k} \right|_{t=0} \right) = \frac{\partial}{\partial x_k} \left(\left. \frac{\partial (\phi_t(x))_j}{\partial t} \right|_{t=0} \right) = \frac{\partial}{\partial x_k} \left(f_j(\phi_t(x)) \right|_{t=0} \right) = \frac{\partial}{\partial x_k} f_j(x).$$

Therefore, we have

$$\left. \frac{d}{dt} \text{vol}(\phi_t(\Lambda)) \right|_{t=0} = \int_{\Lambda} \text{tr} \left(\left. \frac{\partial \mathcal{J}_{\phi_t}(x)}{\partial t} \right|_{t=0} \right) \, dx = \int_{\Lambda} \sum_{k=1}^n \frac{\partial}{\partial x_k} f_k(x) \, dx = \int_{\Lambda} \text{div}(f)(x) \, dx = 0.$$

Now, we replace Λ by $\phi_s(\Lambda)$ for any $s \in (-\epsilon, \epsilon)$ to obtain

$$0 = \left. \frac{d}{dt} \text{vol}(\phi_t(\phi_s(\Lambda))) \right|_{t=0} = \left. \frac{d}{dt} \text{vol}(\phi_{t+s}(\Lambda)) \right|_{t=0} = \left. \frac{d}{dt} \text{vol}(\phi_t(\Lambda)) \right|_{t=s}.$$

Since s is arbitrary, the derivative of $\text{vol}(\phi_t(\Lambda))$ vanishes identically for all $t \in (-\epsilon, \epsilon)$. Thus, we have $\text{vol}(\phi_t(\Lambda)) = \text{vol}(\phi_0(\Lambda)) = \text{vol}(\Lambda)$ for all $t \in (-\epsilon, \epsilon)$. Since preimages under ϕ_t is also images under the inverse ϕ_{-t} , each map ϕ_t is volume-preserving for all $t \in (-\epsilon, \epsilon)$. $\square$

2 Poincaré Recurrence Theorem

For some autonomous first-order differential systems, there exists a bounded set Ω of initial conditions from which all orbits stay inside this set. In other words, Ω is an invariant set. In this case, the flow maps $\phi_t : \Omega \rightarrow \Omega$ can be defined for all $t \in \mathbb{R}$. Indeed, if we initially have a family of maps $\{\phi_t : \Omega \rightarrow \Omega\}_{t \in (-\epsilon, \epsilon)}$ for some $\epsilon > 0$, then by taking compositions of these maps repeatedly, we can define a larger family of maps $\{\phi_t : \Omega \rightarrow \Omega\}_{t \in (-N\epsilon, N\epsilon)}$ for any $N \in \mathbb{N}$. By taking $N \rightarrow \infty$, we can define a flow $\{\phi_t : \Omega \rightarrow \Omega\}_{t \in \mathbb{R}}$. In this case, the system is said to have *bounded orbits*.

All statements and proofs in this section are based on statements given in [1] except Theorem 4 and Theorem 6, which are written with inspiration taken from [2]. For all following statements, a *domain* is a non-empty, connected, open or closed subset of $\mathbb{R}^n$.

Theorem 4 (Poincaré Recurrence Theorem, Weak Form). *Let Ω be a domain of finite volume in $\mathbb{R}^n$ and $\phi : \Omega \rightarrow \Omega$ be a volume-preserving map. Then, for any $x \in \Omega$ and any open neighborhood $\mathcal{N} \subset \Omega$ of x , there exists a $m \in \mathbb{N}$ such that*

$$\phi^m(\mathcal{N}) \cap \mathcal{N} \neq \emptyset,$$

Additionally, the set $\Lambda = \{y \in \mathcal{N} \mid \phi^m(y) \notin \mathcal{N}, \forall m \in \mathbb{N}\}$ is of measure zero.

Proof. By definition, $\phi^m(\Lambda) \cap \mathcal{N} = \emptyset$ for all $m \in \mathbb{N}$. Since $\Lambda \subset \mathcal{N}$, we get $\phi^m(\Lambda) \cap \Lambda = \emptyset$ for all $m \in \mathbb{N}$. For the next step, we use the fact that preimage of an intersection is an intersection of preimages, i.e. $\phi^{-1}(A \cap B) = \phi^{-1}(A) \cap \phi^{-1}(B)$ for any sets $A, B \subset \Omega$. Then, we have for any $m, k \in \mathbb{N}$

$$\emptyset = \phi^{-k}(\phi^m(\Lambda) \cap \Lambda) = \phi^{-k}(\phi^m(\Lambda)) \cap \phi^{-k}(\Lambda) = \phi^{m-k}(\Lambda) \cap \phi^{-k}(\Lambda).$$

Hence, for any $m \in \mathbb{N}$, the sets in $\{\phi^{-k}(\Lambda)\}_{1 \leq k \leq m}$ are mutually disjoint. Since $\bigcup_{k=1}^m \phi^{-k}(\Lambda) \subset \Omega$ and ϕ is volume-preserving, we have for any $m \in \mathbb{N}$

$$\text{vol}(\Omega) \geq \text{vol}\left(\bigcup_{k=1}^m \phi^{-k}(\Lambda)\right) = \sum_{k=1}^m \text{vol}(\phi^{-k}(\Lambda)) = m \cdot \text{vol}(\Lambda).$$

If $\text{vol}(\Lambda) \neq 0$, this shows that $\text{vol}(\Omega)$ is unbounded as $m \rightarrow \infty$ and contradicts the assumption that Ω has a finite volume. Therefore, we must have $\text{vol}(\Lambda) = 0$. Now, we have

$$\text{vol}(\mathcal{N} \setminus \Lambda) = \text{vol}(\mathcal{N}) - \text{vol}(\Lambda) = \text{vol}(\mathcal{N}) > 0,$$

where we used the fact that $\mathcal{N}$ is an open neighborhood of x and thus has a positive volume (it contains an open ball $\mathcal{B}_r(x)$ for some $r > 0$). Hence, $\mathcal{N} \setminus \Lambda$ must be non-empty, i.e. there exist a point $y \in \mathcal{N}$ and $m \in \mathbb{N}$ such that $\phi^m(y) \in \mathcal{N}$. As a result, we have $\phi^m(\mathcal{N}) \cap \mathcal{N} \neq \emptyset$. $\square$

For any autonomous first-order differential system with bounded orbits and a volume-preserving flow, we can set $\phi = \phi_\delta$ for any arbitrary $\delta > 0$. Then, the weak form of Poincaré Recurrence Theorem shows that almost all points in any neighborhood of a point x return to this neighborhood in the future after a finite time $m\delta$. Similarly, if we take $\phi = \phi_{-\delta}$, then almost all points in any neighborhood of a point x start from this neighborhood in a finite time $m\delta$ to the past.

One corollary of this theorem when the map ϕ is invertible with a measurable inverse is as follows.

Corollary 5. *Let Ω be a domain of finite volume in $\mathbb{R}^n$ and $\phi : \Omega \rightarrow \Omega$ be an invertible volume-preserving map with a measurable inverse ϕ^{-1} . Then, for any $x \in \Omega$ there exists a sequence of points $(x_k)_{k \in \mathbb{N}} \subset \Omega$ and a sequence of positive integers $(m_k)_{k \in \mathbb{N}}$ tending to ∞ such that*

$$\lim_{k \rightarrow \infty} x_k = x, \quad \text{and} \quad \lim_{k \rightarrow \infty} \phi^{m_k}(x_k) = x.$$

Proof. From the remark (iii) below Definition 1, the map ϕ^m for any $m \in \mathbb{Z}$ is volume-preserving. Let $m_1 = 1$. We construct the sequences $(x_k)_{k \in \mathbb{N}}$ and $(m_k)_{k \in \mathbb{N}}$ inductively. Suppose that we have chosen m_{k-1} for some $k \geq 2$. By applying Theorem 4 for the volume-preserving map $\phi^{-2m_{k-1}}$ instead of ϕ , there exists a positive integer $m \in \mathbb{N}$ such that

$$\phi^{-2mm_{k-1}}(\mathcal{B}_{1/k}(x)|_\Omega) \cap \mathcal{B}_{1/k}(x)|_\Omega = (\phi^{-2m_{k-1}})^m(\mathcal{B}_{1/k}(x)|_\Omega) \cap \mathcal{B}_{1/k}(x)|_\Omega \neq \emptyset$$

where $\mathcal{B}_{1/k}(x)|_\Omega = \mathcal{B}_{1/k}(x) \cap \Omega$. Then, we set $m_k = 2mm_{k-1}$ and choose an element $x_k \in \phi^{-m_k}(\mathcal{B}_{1/k}(x)|_\Omega) \cap \mathcal{B}_{1/k}(x)|_\Omega$. By construction, we have

$$m_k = 2mm_{k-1} \geq 2m_{k-1} \geq 2^2m_{k-2} \geq \dots \geq 2^k m_0 = 2^k \Rightarrow m_k \xrightarrow{k \rightarrow \infty} \infty.$$

Since $x_k \in \mathcal{B}_{1/k}(x)|_\Omega$, we have

$$\|x - x_k\| < \frac{1}{k} \Rightarrow x_k \xrightarrow{k \rightarrow \infty} x.$$

Since $x_k \in \phi^{-m_k}(\mathcal{B}_{1/k}(x)|_\Omega)$ or $\phi^{m_k}(x_k) \in \mathcal{B}_{1/k}(x)|_\Omega$, we have

$$\|x - \phi^{m_k}(x_k)\| < \frac{1}{k} \Rightarrow \phi^{m_k}(x_k) \xrightarrow{k \rightarrow \infty} x. \quad \square$$

Note that Theorem 4 can be strengthened as follows.

Theorem 6 (Poincaré Recurrence Theorem, Strong Form). *Let Ω be a domain of finite volume in $\mathbb{R}^n$ and $\phi : \Omega \rightarrow \Omega$ be a volume-preserving map. Then, for any $x \in \Omega$ and any neighborhood $\mathcal{N} \subset \Omega$ of x , the set $\Lambda_\infty = \{y \in \mathcal{N} \mid \exists k \geq 0 : \phi^m(y) \notin \mathcal{N}, \forall m > k\}$ is of measure zero.*

Proof. To prove this theorem, we decompose the set Λ_∞ as a countable union of sets according to the largest number k of iterations after which the orbits leave the neighborhood $\mathcal{N}$. First, the set Λ defined in Theorem 4 is exactly the set

$$\Lambda_0 = \{y \in \mathcal{N} \mid \phi^m(y) \notin \mathcal{N}, \forall m > 0\}.$$

For any $k \in \mathbb{N}$, we define the set

$$\Lambda_k = \phi^{-k}(\Lambda_0) \cap \mathcal{N} = \{y \in \mathcal{N} \mid \phi^m(y) \notin \mathcal{N}, \forall m > k\},$$

where the second equality follows from the fact that $y \in \phi^{-k}(\Lambda_0)$ or $\phi^k(y) \in \Lambda_0$ implies that $\phi^m(\phi^k(y)) = \phi^{m+k}(y) \notin \mathcal{N}$ for all $m > 0$. Then, we have

$$\Lambda_\infty = \bigcup_{k=0}^{\infty} \Lambda_k.$$

Using the subadditivity of the Lebesgue measure and the fact that $\Lambda_k \subset \phi^{-k}(\Lambda_0)$, we have

$$\text{vol}(\Lambda_\infty) \leq \sum_{k=0}^{\infty} \text{vol}(\Lambda_k) \leq \sum_{k=0}^{\infty} \text{vol}(\phi^{-k}(\Lambda_0)).$$

In addition, since $\phi : \Omega \rightarrow \Omega$ is volume-preserving, we have $\text{vol}(\phi^{-k}(\Lambda_0)) = \text{vol}(\Lambda_0)$ for all $k \in \mathbb{N}$. Therefore, we get

$$\text{vol}(\Lambda_\infty) \leq \sum_{k=0}^{\infty} \text{vol}(\Lambda_0) = 0,$$

where we used Theorem 4 to get $\text{vol}(\Lambda_0) = 0$. As a result, we must have $\text{vol}(\Lambda_\infty) = 0$. $\square$

This stronger version of the Poincaré Recurrence Theorem shows that for a first-order differential system with bounded orbits, almost all points in any neighborhood of any point return to this neighborhood infinitely many times. Additionally, the Poincaré Recurrence Theorem also has a further implication about the density of initial points whose orbits return arbitrarily close to them infinitely many times. Those such points are defined as follows.

Definition 7. Let Ω be a domain and $\phi : \Omega \rightarrow \Omega$ be a continuous map. A point $x \in \Omega$ is said to be *positively recurrent* with respect to ϕ if there exists a sequence of positive integers $\{m_k\}_{k \in \mathbb{N}}$ tending to ∞ such that $\phi^{m_k}(x) \rightarrow x$. If ϕ is invertible with a continuous inverse ϕ^{-1} , then $x \in \Omega$ is said to be *negatively recurrent* if it is positively recurrent with respect to ϕ^{-1} and *recurrent* if it is both positively and negatively recurrent.

Theorem 8. Let Ω be a closed domain of finite volume in $\mathbb{R}^n$ and $\phi : \Omega \rightarrow \Omega$ be an invertible, continuous, and volume-preserving map with a continuous inverse ϕ^{-1} . Then, the set of recurrent points for ϕ is dense in Ω .

Proof. First of all, in order to prove that the set of positively recurrent points in Ω is dense, we use the assumption that ϕ is invertible, continuous, and volume-preserving with a measurable inverse (continuous maps are measurable maps). Given $\epsilon > 0$ and $N \in \mathbb{N}$, a point $x \in \Omega$ is said to be (ϵ, N) -recurrent with respect to the map ϕ if there exists an $m > N$ such that $\|\phi^m(x) - x\| < \epsilon$. We can show that the set of (ϵ, N) -recurrent points is dense in Ω for any $\epsilon > 0$ and $N \in \mathbb{N}$. Indeed, for any $x \in \Omega$ and $\delta > 0$, we find a (ϵ, N) -recurrent point y such that $\|y - x\| < \delta$ as follows. From Corollary 5, there exist sequences $(x_k)_{k \in \mathbb{N}} \subset \Omega$ and $(m_k)_{k \in \mathbb{N}} \subset \mathbb{N}$ such that $x_k \rightarrow x$, $m_k \rightarrow \infty$, and $\phi^{m_k}(x_k) \rightarrow x$ as $k \rightarrow \infty$. Hence, there exist $k_1, k_2, k_3 \in \mathbb{N}$ such that

$$\begin{aligned} \|x_k - x\| &< \min\{\delta, \epsilon/2\} \quad \forall k \geq k_1, & m_k &> N \quad \forall k \geq k_2, \\ \|\phi^{m_k}(x_k) - x\| &< \epsilon/2 \quad \forall k \geq k_3. \end{aligned}$$

By choosing $k_0 = \max\{k_1, k_2, k_3\}$, we get a desired (ϵ, N) -recurrent point $y = x_{k_0}$ since

$$\begin{aligned} \|x_{k_0} - x\| &< \delta, & m_{k_0} &> N, \\ \|\phi^{m_{k_0}}(x_{k_0}) - x_{k_0}\| &\leq \|x - x_{k_0}\| + \|\phi^{m_{k_0}}(x_{k_0}) - x\| < \epsilon/2 + \epsilon/2 = \epsilon. \end{aligned}$$

In addition, the set of (ϵ, N) -recurrent points is also open in Ω since ϕ is continuous and thus the map $g : \Omega \ni x \mapsto \|\phi^m(x) - x\| \in \mathbb{R}$ is continuous. Therefore, for any (ϵ, N) -recurrent point x and $\epsilon' \in (0, \epsilon - g(x))$, there exists $\delta > 0$ such that for any $y \in \Omega$ with $\|y - x\| < \delta$, we have

$$|g(y) - g(x)| < \epsilon' \quad \Rightarrow \quad \|\phi^m(y) - y\| = g(y) < g(x) + \epsilon' < \epsilon,$$

which shows that y is also (ϵ, N) -recurrent.

Now, we observe that any positively recurrent point is $(1/N, N)$ -recurrent for all $N \in \mathbb{N}$. Indeed, if x is $(1/N, N)$ -recurrent for all $N \in \mathbb{N}$, then we can choose an integer $m_N > N$ for each N so that $\|\phi^{m_N}(x) - x\| < 1/N$, which implies that $m_N \rightarrow \infty$ and $\phi^{m_N}(x) \rightarrow x$ as $N \rightarrow \infty$. Conversely, if x is a positively recurrent, then there exists $\{m_k\}_{k \in \mathbb{N}}$ tending to ∞ such that $\phi^{m_k} \rightarrow x$ as $k \rightarrow \infty$. Then, for any $N \in \mathbb{N}$, there exists k_N such that $m_k > N$ and $\|\phi^{m_k}(x) - x\| < 1/N$ for any $k \geq k_N$, which implies that x is $(1/N, N)$ -recurrent for all $N \in \mathbb{N}$. Hence, the set of positively recurrent points is a countable intersection of all sets of $(1/N, N)$ -recurrent points for all $N \in \mathbb{N}$. By Baire category theorem, the intersection of countably many dense open sets is dense. Thus, the set of positively recurrent points is dense in Ω .

Similarly, by applying the same argument for the map ϕ^{-1} , we can show that the set of negatively recurrent points is also dense in Ω . Therefore, the set of recurrent points being the intersection of two dense sets is also dense in Ω . $\square$

3 Autonomous Hamiltonian systems

Consider a function $\mathbb{R}^n \times \mathbb{R}^n \ni (q, p) \mapsto \mathcal{H}(q, p) \in \mathbb{R}$ of class C^2 and a differential system $\dot{x} = f(x)$ where $x(t) = (q(t), p(t))^t$ and $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ given by

$$f = \left(\frac{\partial \mathcal{H}}{\partial p_1}, \dots, \frac{\partial \mathcal{H}}{\partial p_n}, -\frac{\partial \mathcal{H}}{\partial q_1}, \dots, -\frac{\partial \mathcal{H}}{\partial q_n} \right)^t.$$

Such a system is called an *autonomous Hamiltonian system* with the *Hamiltonian* $\mathcal{H}$. The vector q represents a point in a *configuration space*, while the vector x represents a point in

a *phase space*. Additionally, q and p are also called *coordinate* and *momentum*, respectively. Since $\mathcal{H}$ is of class C^2 , the vector field f is differentiable and we have

$$\operatorname{div}(f) = \sum_{i=1}^n \frac{\partial}{\partial q_i} \left(\frac{\partial \mathcal{H}}{\partial p_i} \right) + \sum_{i=1}^n \frac{\partial}{\partial p_i} \left(-\frac{\partial \mathcal{H}}{\partial q_i} \right) = \sum_{i=1}^n \left(\frac{\partial^2 \mathcal{H}}{\partial q_i \partial p_i} - \frac{\partial^2 \mathcal{H}}{\partial p_i \partial q_i} \right) = 0,$$

which shows that the flow of a Hamiltonian system is volume-preserving by Proposition 3. This result is known as [Liouville's theorem](#) in Hamiltonian mechanics. Moreover, one useful fact about this system is that $\mathcal{H}(q(t), p(t))$ is a constant function of time t for any solution $x(t) = (q(t), p(t))^t$ of the system. This can be seen from

$$\frac{d}{dt} \mathcal{H}(q(t), p(t)) = \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial q_i} \dot{q}_i(t) + \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial p_i} \dot{p}_i(t) = \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial q_i} \left(\frac{\partial \mathcal{H}}{\partial p_i} \right) + \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial p_i} \left(-\frac{\partial \mathcal{H}}{\partial q_i} \right) = 0.$$

This fact has an important implication for any autonomous Hamiltonian system. That is, every orbit of this system stays on a $(2n - 1)$ -dimensional hypersurface in the phase space. Next, we consider two well-known examples of autonomous Hamiltonian systems.

Example 9. A one-dimensional harmonic oscillator is a Hamiltonian system with the Hamiltonian $\mathcal{H} : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$\mathcal{H}(q, p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2,$$

where $m, \omega > 0$ are the mass and the angular frequency of the oscillator, respectively. The corresponding differential system is given by

$$\begin{pmatrix} \dot{q} \\ \dot{p} \end{pmatrix} = \begin{pmatrix} \frac{\partial \mathcal{H}}{\partial p} (q, p) \\ -\frac{\partial \mathcal{H}}{\partial q} (q, p) \end{pmatrix} = \begin{pmatrix} \frac{p}{m} \\ -m\omega^2 q \end{pmatrix}.$$

The first equation gives $p = m\dot{q}$ and substituting it into the second equation yields

$$m\ddot{q} = -m\omega^2 q \quad \Leftrightarrow \quad \ddot{q} + \omega^2 q = 0,$$

which has the general solution

$$q(t) = A \cos(\omega t) + B \sin(\omega t)$$

for some constants $A, B \in \mathbb{R}$. The corresponding solution for $p(t)$ is given by

$$p(t) = m\dot{q}(t) = -m\omega A \sin(\omega t) + m\omega B \cos(\omega t).$$

Consider the initial conditions $q(0) = q_0$ and $p(0) = p_0$. Then, we obtain a unique solution

$$\begin{cases} q(t) = q_0 \cos(\omega t) + \frac{p_0}{m\omega} \sin(\omega t), \\ p(t) = p_0 \cos(\omega t) - m\omega q_0 \sin(\omega t). \end{cases}$$

Therefore, the flow $\phi_t : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ of the system is a linear map given by

$$\phi_t \begin{pmatrix} q_0 \\ p_0 \end{pmatrix} = \begin{pmatrix} \cos(\omega t) & \frac{1}{m\omega} \sin(\omega t) \\ -m\omega \sin(\omega t) & \cos(\omega t) \end{pmatrix} \begin{pmatrix} q_0 \\ p_0 \end{pmatrix}.$$

It is clear that ϕ_t is periodic with the period $T = \frac{2\pi}{\omega}$, i.e., $\phi_{t+T} = \phi_t$ for all $t \in \mathbb{R}$. The orbits of this system are closed ellipses centered at the origin as shown in Figure 1. Thus, every point in $\mathbb{R}^2$ is recurrent with respect to the map $\phi_{T/n}$ for any $n \geq 1$.

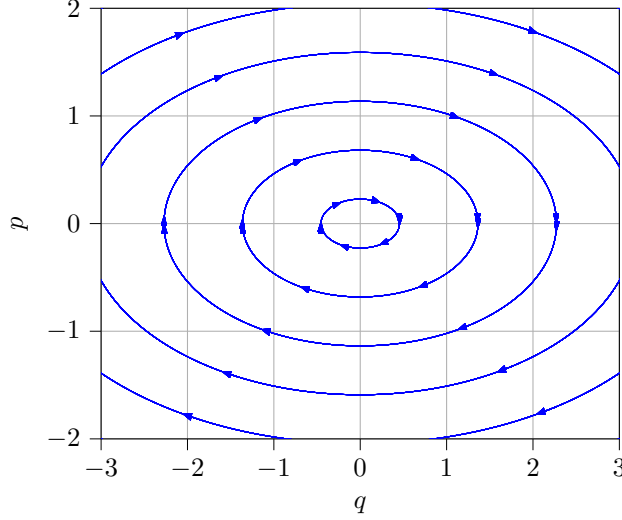


Figure 1: Integral curves of $F(q, p) = (p/m, -m\omega^2 q)$ for $m = 1$ and $\omega = 0.5$.

Example 10. A Kepler problem describes an autonomous Hamiltonian system given by the Hamiltonian $\mathcal{H} : \mathbb{R}^2 \setminus \{0\} \times \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$\mathcal{H}(q, p) = \frac{\|p\|^2}{2m} - \frac{k}{\|q\|} = \frac{p_1^2 + p_2^2}{2m} - \frac{k}{\sqrt{q_1^2 + q_2^2}},$$

where $m, k > 0$ are constants. The corresponding differential system is given by

$$\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{p}_1 \\ \dot{p}_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial \mathcal{H}}{\partial p_1}(q, p) \\ \frac{\partial \mathcal{H}}{\partial p_2}(q, p) \\ -\frac{\partial \mathcal{H}}{\partial q_1}(q, p) \\ -\frac{\partial \mathcal{H}}{\partial q_2}(q, p) \end{pmatrix} = \begin{pmatrix} \frac{p_1}{m} \\ \frac{p_2}{m} \\ -\frac{kq_1}{(q_1^2 + q_2^2)^{3/2}} \\ -\frac{kq_2}{(q_1^2 + q_2^2)^{3/2}} \end{pmatrix}.$$

The first two equations give $p = m\dot{q}$ and substituting it into the last two equations yields

$$m\ddot{q} = -\frac{kq}{\|q\|^3} \Leftrightarrow \ddot{q} = -\frac{kq}{m\|q\|^3}. \quad (\star)$$

To solve this equation, we use polar coordinates $q = (r \cos \theta \ r \sin \theta)^t$ for $r = \|q\| > 0$ and $\theta \in \mathbb{R}$. Solving for $r(t)$ and $\theta(t)$ is equivalent to solving for $q(t)$. First, we compute

$$\begin{aligned} \dot{q} &= \begin{pmatrix} \dot{r} \cos \theta - r \dot{\theta} \sin \theta \\ \dot{r} \sin \theta + r \dot{\theta} \cos \theta \end{pmatrix} = \dot{r} e_r + r \dot{\theta} e_\theta, \\ \ddot{q} &= \begin{pmatrix} (\ddot{r} - r \dot{\theta}^2) \cos \theta - (2\dot{r} \dot{\theta} + r \ddot{\theta}) \sin \theta \\ (\ddot{r} - r \dot{\theta}^2) \sin \theta + (2\dot{r} \dot{\theta} + r \ddot{\theta}) \cos \theta \end{pmatrix} = (\ddot{r} - r \dot{\theta}^2) e_r + (2\dot{r} \dot{\theta} + r \ddot{\theta}) e_\theta, \end{aligned}$$

where $e_r = (\cos \theta \ \sin \theta)^t = q/\|q\|$ and $e_\theta = (-\sin \theta \ \cos \theta)^t$ are two orthonormal vectors. Projecting both sides of $(\star)$ onto e_r and e_θ , we obtain the system

$$\begin{cases} \ddot{r} - r \dot{\theta}^2 = -\frac{k}{mr^2}, \\ 2\dot{r} \dot{\theta} + r \ddot{\theta} = 0, \end{cases} \Leftrightarrow \begin{cases} \ddot{r} - r \dot{\theta}^2 = -\frac{k}{mr^2}, \\ \frac{1}{r} \frac{d}{dt}(r^2 \dot{\theta}) = 0. \end{cases}$$

The second equation implies that $r^2\dot{\theta} = l/m$ for some constant $l \in \mathbb{R}$. The constant l is called the angular momentum. Substituting $\dot{\theta} = l/(mr^2)$ into the first equation yields

$$\ddot{r} - \frac{l^2}{m^2 r^3} = -\frac{k}{mr^2}. \quad (\star\star)$$

In the case $l = 0$, we have $p \cdot e_\theta = m\dot{q} \cdot e_\theta = m\dot{\theta} = 0$ and $\theta = \text{const.}$ Therefore, we only need to consider

$$\begin{cases} \dot{r} = p_r/m, \\ \dot{p}_r = -k/r^2, \end{cases}$$

where $p_r = p \cdot e_r = m\dot{r}$ and $\dot{p}_r = m\ddot{r}$. This system has solutions which are unbounded in either r or p_r (see Figure 2).

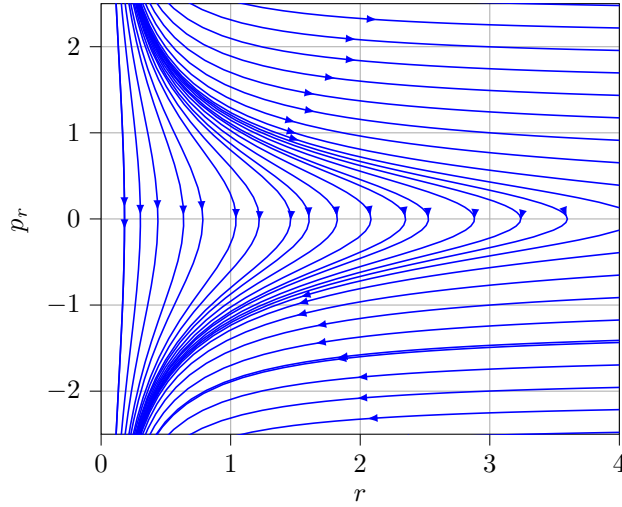


Figure 2: Integral curves of $F(r, p_r) = (p_r, -1/r^2)$ for $r > 0$.

Now, suppose that $l \neq 0$, which means that $\dot{\theta} \neq 0$ and thus $t \mapsto \theta(t)$ is strictly monotonic. Therefore, we can invert $t \mapsto \theta(t)$ to obtain $t = t(\theta)$ and define $r(\theta) = r(t(\theta))$. Using the chain rule, we have

$$\dot{r} = \frac{dr}{d\theta} \dot{\theta} = \frac{dr}{d\theta} \frac{l}{mr^2}, \quad \ddot{r} = \frac{d}{d\theta} \left(\frac{dr}{d\theta} \frac{l}{mr^2} \right) \dot{\theta} = \left[\frac{d^2r}{d\theta^2} \frac{l}{mr^2} - \frac{2l}{mr^3} \left(\frac{dr}{d\theta} \right)^2 \right] \frac{l}{mr^2}.$$

Now, let $u(\theta) = 1/r(\theta)$. Then, we have

$$\frac{dr}{d\theta} = -\frac{1}{u^2} \frac{du}{d\theta}, \quad \frac{d^2r}{d\theta^2} = \frac{2}{u^3} \left(\frac{du}{d\theta} \right)^2 - \frac{1}{u^2} \frac{d^2u}{d\theta^2}.$$

Using these expressions, we obtain

$$\begin{aligned} \ddot{r} &= \left\{ \left[\frac{2}{u^3} \left(\frac{du}{d\theta} \right)^2 - \frac{1}{u^2} \frac{d^2u}{d\theta^2} \right] \frac{l}{m} u^2 - \frac{2l}{m} u^3 \left(-\frac{1}{u^2} \frac{du}{d\theta} \right)^2 \right\} \frac{l}{m} u^2 \\ &= -\frac{l^2}{m^2} u^2 \frac{d^2u}{d\theta^2}. \end{aligned}$$

Substituting this expression into $(\star\star)$, we have

$$-\frac{l^2}{m^2} u^2 \frac{d^2u}{d\theta^2} - \frac{l^2}{m^2} u^3 = -\frac{k}{m} u^2 \quad \Leftrightarrow \quad \frac{d^2u}{d\theta^2} + u = \frac{km}{l^2}, \quad (\star\star\star)$$

where we have used the fact that $u = 1/r \neq 0$. The general solution of this equation is given by

$$u(\theta) = \frac{km}{l^2} + A \cos(\theta - \theta_0)$$

for some constants $A \geq 0$ and $\theta_0 \in \mathbb{R}$. Therefore, we have

$$r(\theta) = \frac{1}{u(\theta)} = \frac{r_0}{1 + e \cos(\theta - \theta_0)},$$

where $r_0 = l^2/(km)$ and $e = Al^2/(km)$, called the eccentricity. Now, recalling that $H(q(t), p(t))$ is constant, denoted by E , in t for any solution $q(t), p(t)$, we compute

$$\begin{aligned} E = \mathcal{H}(q, p) &= \frac{\|p\|^2}{2m} - \frac{k}{\|q\|} = \frac{m\|\dot{q}\|^2}{2} - \frac{k}{r} = \frac{m}{2} (\dot{r}^2 + r^2\dot{\theta}^2) - \frac{k}{r(t)} = \frac{m}{2} \dot{r}^2 + \frac{l^2}{2mr^2} - \frac{k}{r} \\ &= \frac{m}{2} \left(-\frac{l}{m} \frac{du}{d\theta} \right)^2 + \frac{l^2}{2m} u^2 - ku = \frac{l^2}{2m} \left[\left(\frac{du}{d\theta} \right)^2 + \left(u - \frac{km}{l^2} \right)^2 \right] - \frac{mk^2}{2l^2} = \frac{l^2}{2m} A^2 - \frac{mk^2}{2l^2}. \end{aligned}$$

Using this expression, we obtain

$$e = \frac{l^2}{km} A = \sqrt{1 + \frac{2El^2}{mk^2}}.$$

Based on this expression, we have three cases:

- $e < 1$ or $E < 0$: the orbit in configuration space is an ellipse;
- $e = 1$ or $E = 0$: the orbit in configuration space is a parabola;
- $e > 1$ or $E > 0$: the orbit in configuration space is a hyperbola.

Therefore, the orbit in configuration space is closed and periodic when $E < 0$. Consider the set of initial points in phase space given by

$$\Omega = \{(q, p) \in \mathbb{R}^2 \setminus \{0\} \times \mathbb{R}^2 \mid \mathcal{H}(q, p) < 0, p \cdot e_\theta \neq 0\}.$$

Every point $(q, p) \in \Omega$ is recurrent with respect to the map $\phi_{T/n}$ for any $n \geq 1$, where T is the period of the function $t \mapsto \phi_t(q, p)$. However, observe that the domain Ω is not bounded and hence, it does not have a finite volume since $\|p\|^2/2m = E + k/\|q\|$ becomes arbitrarily large as $\|q\|$ approaches 0.

The two examples that we have considered so far are special because they have periodic solutions. Therefore, we do not need a domain of finite volume to have recurrent points. To conclude this report, we consider a further example of an autonomous Hamiltonian system which has bounded and periodic orbits.

Example 11. Consider a system with the Hamiltonian $\mathcal{H} : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$\mathcal{H}(q, p) = \sqrt{p^2 c^2 + m^2 c^4} + \frac{1}{2} m \omega^2 q^2 + \frac{\beta}{4} q^4,$$

where m, ω, β, c are positive constants. This system describes a relativistic particle moving in a potential well given by $V(q) = \frac{1}{2} m \omega^2 q^2 + \frac{\beta}{4} q^4$. The corresponding differential system is given by

$$\begin{pmatrix} \dot{q} \\ \dot{p} \end{pmatrix} = \begin{pmatrix} \frac{\partial \mathcal{H}}{\partial p}(q, p) \\ -\frac{\partial \mathcal{H}}{\partial q}(q, p) \end{pmatrix} = \begin{pmatrix} \frac{pc^2}{\sqrt{p^2 c^2 + m^2 c^4}} \\ -m\omega^2 q - \beta q^3 \end{pmatrix}.$$

This system cannot be solved analytically in general. However, we can solve it numerically for some initial conditions $q(0) = q_0$ and $p(0) = p_0$ (see Figure 3). From the figure, we observe that the orbits are bounded and periodic. The boundedness of the orbits can be seen from the fact that $\mathcal{H}(q(t), p(t)) = E$ for any t where $E > 0$ is some constant, and

$$E = \sqrt{p(t)^2 c^2 + m^2 c^4} + \frac{1}{2} m \omega^2 q(t)^2 + \frac{\beta}{4} q(t)^4 \Rightarrow \begin{cases} E > \frac{\beta}{4} q(t)^4, \\ E > p(t)c. \end{cases}$$

This shows that both $t \mapsto q(t)$ and $t \mapsto p(t)$ have to be bounded.

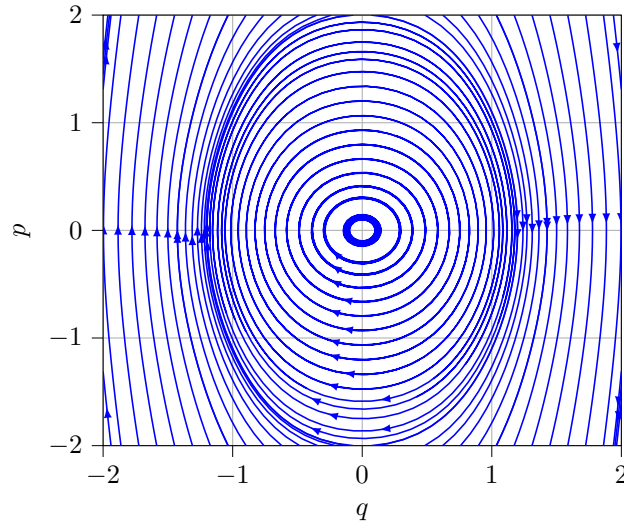


Figure 3: Integral curves of $F(q, p) = (pc^2/\sqrt{p^2c^2 + m^2c^4}, -m\omega^2q - \beta q^3)$ for $m = 1$, $c = 1$, $\omega = 1$, $\beta = 1$.

References

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